

ToU equity lab meeting

Who pays the fees?

Accessibility-based equity of congestion charging scenarios in Auckland

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The research question

Who would face a Time of Use charge in Auckland **without a viable public transport alternative** under each of the six shortlisted scenarios, and how would that burden be distributed against area-level deprivation?

Burden

Who would pay the charge under each scenario?

Alternative

Does that population have a viable 45-minute public transport alternative?

Distribution

How does the burden sit against area-level deprivation?

Congestion pricing is no longer a hypothetical.

The Land Transport (Time of Use Charging) Amendment Act 2025 sets the legal frame. Auckland Transport has six shortlisted scenarios. We do not adjudicate whether pricing should happen, but which option carries which equity cost.

84.7%

of Aucklanders travel by car (Census 2023)

NZ\$6

peak cordon charge under consideration

6

shortlisted scenarios, three families

Three families, six footprints.

1 City centre cordons

1a, CBD only (15)

1c, CBD + fringe (27)

2 Isthmus double cordon

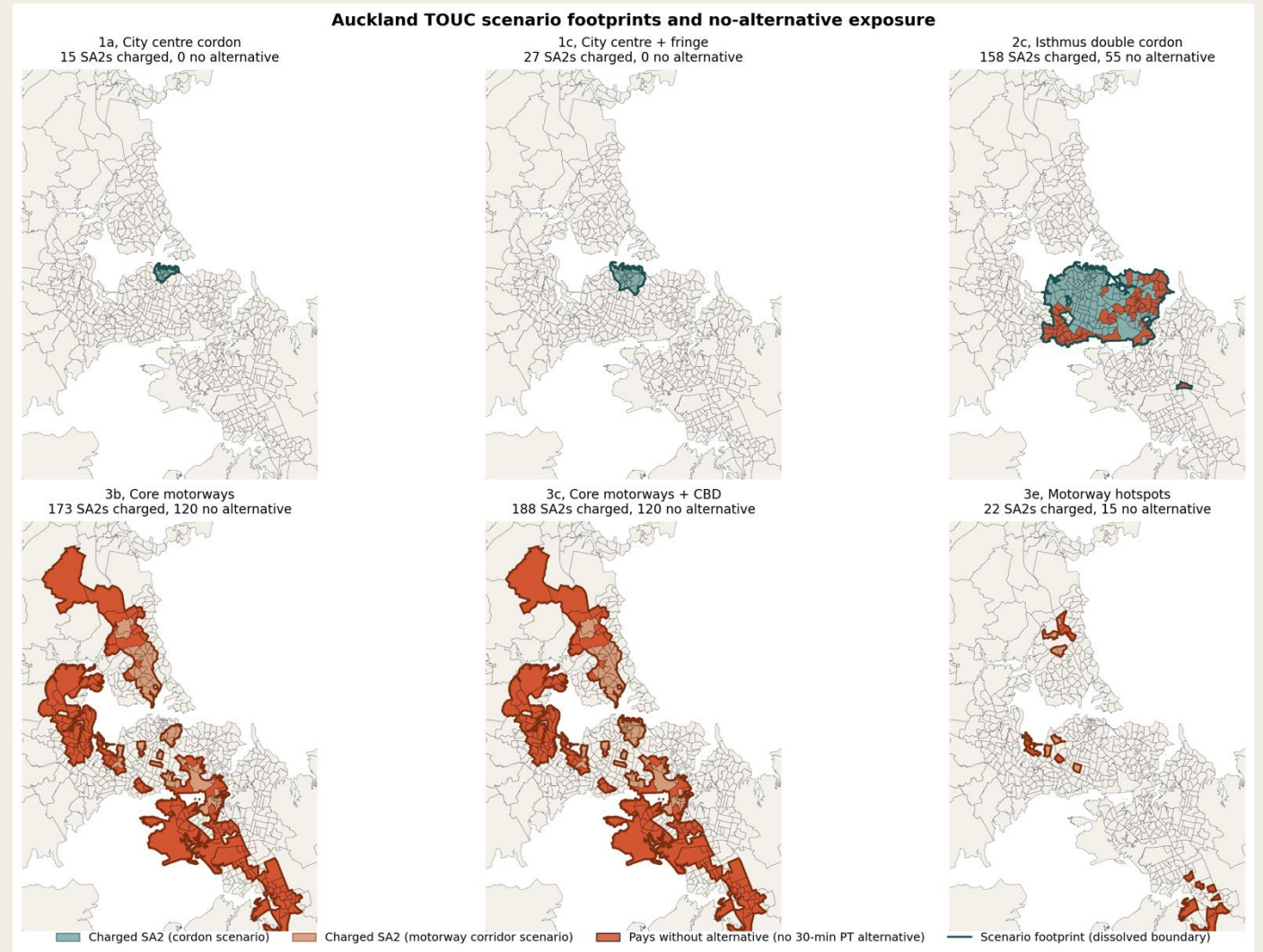
2c, expanded inner isthmus (158)

3 Motorway corridors

3b, core motorways (173)

3c, core motorways + CBD (188)

3e, motorway hotspots (22)



Five steps from raw GTFS to a scenario equity index.

1

Routing

r5py over the Auckland Transport GTFS feed (22 April 2026) and OpenStreetMap walking network. Origins are SA2 population-weighted centroids.

2

Accessibility

Jobs reachable within a 45-minute weekday peak window, 07:00 to 09:00, combined walk and PT. 45 min chosen because Auckland's mean commute is 32 min.

3

Q75 cut-point

Pre-specified within-Auckland Q75 of access_45min, 104,812 jobs. The top quartile is a conservative test for what counts as a viable alternative.

4

Burden class

Each SA2 coded pays_with_alternative, pays_without_alternative, or does_not_pay. Cross-tabulated against NZDep2023 decile.

5

Equity index

Wagstaff Concentration Index of access_45min against NZDep2023, both baseline (619 SA2s) and per-scenario on the charged subset. SA1-population-weighted.

Auckland's access surface is non-monotonic.

Two low-access tails.

Decile 1 is affluent rural fringe (Coatesville, Riverhead, Kumeu, Maraetai). Low access by choice, not constraint.

Decile 10 is the outer south and outer west (Manurewa, Papakura, Henderson, Ranui). Low access as a binding constraint.

Peak at D6-7 averages around 82,000 reachable jobs.



Box plots of access_45min by NZDep decile, with LOWESS trend. Q75 line at 86k jobs (this older run uses an earlier weighting; current table uses 104,812).

Overall CI sits near zero (+0.031) because the two low tails average out. The right answer is the D8-10 subgroup CI of -0.083.

One in five Aucklanders.

20.7%

*of Auckland's 1.65 million people would face the charge under 3b or 3c with no viable
45-minute public transport alternative*

340,908 residents | 120 SA2s | mostly outer south and outer west

TABLE 1, BURDEN INCIDENCE

How many residents pay without an alternative?

45-minute Q75 cut-point = 104,812 reachable jobs. Residents column = SA1-summed Census 2023 usually resident count in 'pays without alternative' SA2s.

Scenario	Charged SA2	Pays without alt SA2	Residents affected	% of Auckland (1.65M)
1a, City centre cordon	15	0	0	0.0%
1c, City centre + fringe	27	0	0	0.0%
2c, Isthmus double cordon	158	57	151,050	9.2%
3b, Core motorways	173	120	340,908	20.7%
3c, Core motorways + CBD	188	120	340,908	20.7%
3e, Motorway hotspots	22	15	28,614	1.7%

TABLE 2, SCENARIO CI

Same six scenarios, different question.

Among the SA2s that would pay, do the more deprived ones have lower 45-minute accessibility? Wagstaff CI restricted to the charged subset, population-weighted by SA1 usually resident count. Baseline (all 619 SA2s) CI(access_45min) = +0.031, D8-10 = -0.083.

Scenario	Charged SA2	Residents in charged SA2	CI(access_45min charged)	Equity reading
1a, City centre cordon	15	31,107	-0.005	<i>not equity-active</i>
1c, City centre + fringe	27	57,420	+0.029	<i>not equity-active</i>
2c, Isthmus double cordon	158	414,066	+0.010	<i>near-zero</i>
3b, Core motorways	173	476,415	-0.166	<i>strongly regressive</i>
3c, Core motorways + CBD	188	507,522	-0.120	<i>regressive</i>
3e, Motorway hotspots	22	51,588	+0.046	<i>near-zero</i>

Adding the CBD compresses the CI but leaves the burden unchanged.

3b

CI -0.166

Core motorways

340,908 residents | 120 no-alt SA2s

3c

CI -0.120

Core motorways + CBD

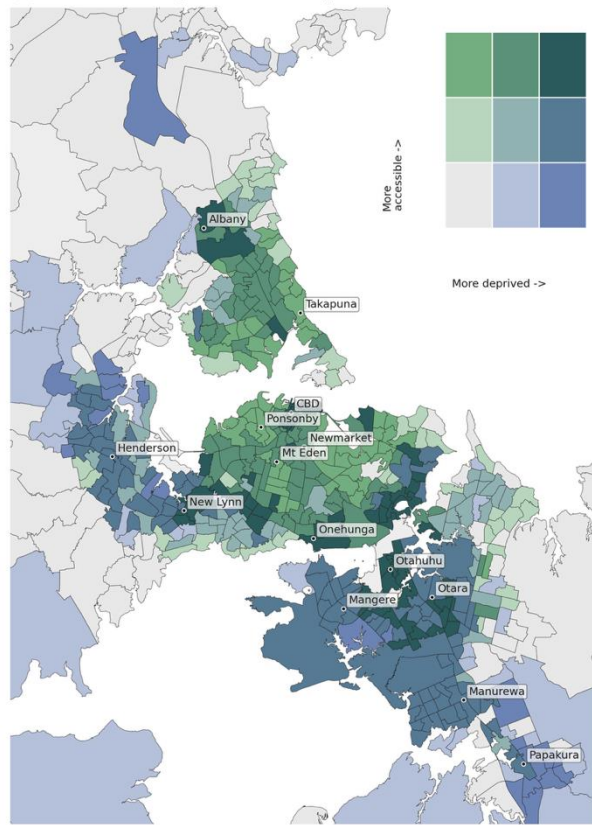
340,908 residents | 120 no-alt SA2s

The same 340,908 residents would pay without a viable 45-minute alternative under both 3b and 3c. The CBD SA2s 3c adds sit well above the Q75 cut-point and span the deprivation gradient, so the regressive gradient is diluted. **A measurement effect, not a real improvement in distributional outcomes.**

Four local clusters tell the policy story.

Bivariate Local Moran's I, access_45min focal, NZDep2023 spatial lag, queen weights, 999 permutations, $p < 0.05$.

(c) Bivariate map: access x deprivation



HH

34 SA2s

High access, high deprivation

Rail-served deprived band of the inner-south, including Point England, Glen Innes, Otahuhu, Mangere Central, parts of Onehunga. The data-driven reason H2 does not classify the deprived inner-south as paying without an alternative.

HL

26 SA2s

High access, low deprivation

Inner-isthmus elite belt, Remuera, Mt Eden, Newmarket, the inner CBD ring.

LH

76 SA2s

Low access, high deprivation

The H2 burden cluster proper, outer Manurewa, Papakura, Takanini, outer Henderson, Massey, and Ranui.

LL

81 SA2s

Low access, low deprivation

Affluent rural and coastal fringe, Coatesville, Riverhead, Whangaparaoa, Kumeu, Maraetai, which produces the decile 1 tail of the inverted-U.

No charge without an alternative.

We take no position on whether Auckland should introduce ToU charging. Given the policy decision has been made under the Land Transport (Time of Use Charging) Amendment Act 2025, three implications follow from the data.

1

Inner cordon options 1a and 1c can proceed

The population paying the charge under those scenarios has a clearly above-average public transport alternative. No distributional preconditions.

2

Motorway options 3b and 3c require a preceding PT step

Credible public transport investment in the affected outer SA2s is needed before the charge takes effect. Otherwise the scheme would charge 340,908 residents for a journey it has not yet given them a way to make.

3

Expanded isthmus 2c is a different question

Non-trivial cost on a sizeable share of Auckland without a clear distributional sign. The policy frame for that scenario is total cost-benefit, not equity.

Three extensions, in priority order.

01

Per-corridor PT audit

Map each of the 120 no-alt SA2s to the specific bus, rail, or active travel intervention that would lift it above Q75 before the charge.

02

SA1-resolution accessibility

Routing from 10,207 SA1 centroids directly, rather than inheriting SA2 access. Reveals within-SA2 heterogeneity.

03

HOP patronage validation

Test the alternative classification against observed HOP growth on recently service-uplifted AT corridors since 2023.

Thanks. Questions welcome.